ANOMALY DETECTION

Complete Study Notes & Concepts

Machine Learning • Statistics • Deep Learning • Real-World Applications

- Introduction & Definitions
- Types of Anomalies
- Proximity-Based Methods
- Evaluation Metrics
- Time-Series Anomaly Detection
- Statistical Methods
- Machine Learning Approaches
- Deep Learning Methods
- Industry Applications
- Challenges & Best Practices

What is Anomaly Detection?

Anomaly Detection (also called Outlier Detection) is the process of identifying data points, observations, or patterns that deviate significantly from the expected behavior of a dataset. These deviations — called anomalies, outliers, or novelties — may signal critical events such as fraud, system failures, medical conditions, or security breaches.

> *Definition: An anomaly is an observation that deviates so much from other observations as to arouse suspicion that it was generated by a different mechanism.*

Why Is It Important?

- Fraud Detection — Credit card fraud, insurance fraud, money laundering
- Cybersecurity — Intrusion detection, malware identification, DDoS attacks
- Healthcare — Disease outbreak detection, patient monitoring, ECG anomalies
- Manufacturing — Defective product detection, predictive maintenance
- Finance — Stock market manipulation, unusual trading patterns
- Network Monitoring — Server downtime, latency spikes, bandwidth anomalies
- IoT & Sensors — Sensor drift, equipment faults, environmental anomalies

Key Terminology

Anomaly / Outlier	A data point that is significantly different from the majority of the data.
Inlier	A data point that conforms to the expected normal behavior.
Novelty	A new unobserved pattern in the data (in novelty detection, training data has no anomalies).
Contamination Rate	The proportion of anomalies present in the dataset (typically 1–10%).
Decision Boundary	The boundary separating normal from anomalous observations.
Anomaly Score	A score assigned to each point indicating how anomalous it is.
Threshold	The cut-off value on the anomaly score above which a point is declared anomalous.

Point Anomaly

A single data instance is anomalous with respect to the rest of the data.

- Most common type of anomaly.
- Example: A single transaction of $10,000 when average is $50.
- Detected by comparing a point's value to the overall distribution.
- Techniques: Z-Score, IQR, Isolation Forest, LOF.

Contextual Anomaly

A data instance is anomalous in a specific context, but not otherwise.

- Also called Conditional Anomaly.
- The same value can be normal in one context and anomalous in another.
- Example: 35°C temperature is normal in summer but anomalous in winter.
- Requires context attributes (e.g., time, location) and behavioral attributes.
- Techniques: Seasonal decomposition, SARIMA, LSTM.

Collective Anomaly

A collection of related data instances is anomalous with respect to the entire dataset.

- Individual points may be normal, but their combination is unusual.
- Example: A sequence of ECG readings forming an abnormal pattern.
- Common in time-series, sequences, and spatial data.
- Techniques: Sequence mining, HMM, LSTM autoencoders.

Anomaly Detection by Data Granularity

Dimension	Description	Example
Global Anomaly	Point is far from entire data distribution	Data value 3 std devs from mean
Local Anomaly	Point is far from its local neighborhood	Dense cluster surrounds a slightly distant point
Group Anomaly	Subset of points is anomalous collectively	A batch of similar sensor readings

High-Level Taxonomy

Category	Training Label	Assumption	Common Methods
Supervised	Requires labeled normal & anomalous data	Both classes available during training	SVM, Random Forest, Neural Networks
Semi-Supervised	Only normal data labeled	Model learns 'normal'; deviations are anomalous	One-Class SVM, Autoencoders, SVDD
Unsupervised	No labels required	Anomalies are rare and different from normal	K-Means, DBSCAN, Isolation Forest, LOF

Supervised vs Unsupervised Trade-offs

- Supervised methods offer highest accuracy but require expensive labeled data (anomalies are rare).
- Unsupervised methods are widely used in practice due to no labeling requirement.
- Semi-supervised methods are a practical middle ground — label only the normal class.
- Online/Streaming methods update the model in real time as data arrives.

Method Families at a Glance

Method Family	Core Idea	Strengths	Weaknesses
Statistical	Model data distribution; flag low-probability points	Interpretable, fast	Assumes distribution form
Distance-Based	Anomalies are far from their neighbors	No distribution assumption	Slow on large data
Density-Based	Anomalies in low-density regions	Handles clusters of different densities	Sensitive to params
Clustering	Anomalies don't belong to any cluster	Unsupervised, scalable	Cluster shape assumptions
Tree-Based	Anomalies are easier to isolate in trees	Fast, scalable, no distribution needed	Less interpretable
Deep Learning	Learn complex normal representations	Handles high-dim, sequential data	Needs large data, black box

[redacted]

4.1 Z-Score (Standard Score)

Measures how many standard deviations a data point is from the mean. Works best for normally distributed univariate data.

Formula:

```
z = (x - mu) / sigma
```

Where x = observation, mu = mean, sigma = standard deviation.

- Threshold: |z| > 2 (95%) or |z| > 3 (99.7%) flags as anomaly.
- Limitation: Sensitive to the mean and std themselves being affected by outliers.
- Modified Z-Score uses median and MAD for robustness: M = 0.6745 * (x - median) / MAD

4.2 Interquartile Range (IQR) Method

```
IQR = Q3 - Q1
```

```
Lower Bound = Q1 - 1.5 * IQR Upper Bound = Q3 + 1.5 * IQR
```

- Points outside [Lower, Upper] bounds are flagged as anomalies.
- Robust to extreme values — uses quartiles not mean/std.
- Use factor 3.0 instead of 1.5 for extreme outlier detection.

4.3 Grubbs' Test (Extreme Studentized Deviate)

```
G = max|xi - x_bar| / s
```

- Tests the null hypothesis: No outliers in the dataset.
- Assumes normality; tests one outlier at a time (iterative).
- Compare G against critical value from t-distribution.

4.4 Gaussian Mixture Model (GMM)

Models the data as a mixture of K Gaussian distributions. Anomaly score = negative log-likelihood under the fitted model.

```
P(x) = sum_k [ pi_k * N(x | mu_k, Sigma_k) ]
```

- Fitted using Expectation-Maximization (EM) algorithm.
- Points with very low P(x) are considered anomalies.
- Extends to multivariate data naturally.
- Limitation: Assumes data is generated from Gaussian components.

4.5 Kernel Density Estimation (KDE)

Non-parametric density estimation — no assumption about the distribution shape. Estimates the PDF from data using a kernel function (e.g., Gaussian).

```
f_h(x) = (1/n) * sum_i [ K((x - xi) / h) ] / h
```

- h = bandwidth (smoothing parameter) — controls bias-variance trade-off.
- Anomaly score = 1 / f(x). Low-density regions give high anomaly scores.
- Bandwidth selection: Scott's rule, Silverman's rule, cross-validation.

4.6 CUSUM (Cumulative Sum Control Chart)

Designed for detecting small, persistent shifts in the mean of sequential data.

```
S_t+ = max(0, S_(t-1)+ + x_t - mu_0 - k) [upper CUSUM] S_t- = max(0, S_(t-1)- - x_t +
mu_0 - k) [lower CUSUM]
```

- k = slack (allowance); h = decision threshold.
- Signal when S_t > h (change detected).
- Optimal for detecting step-changes in time series.

5.1 k-Nearest Neighbor (kNN) Anomaly Detection

Anomaly score of a point is based on its distance to the k-th nearest neighbor. Points far from their neighbors are likely anomalies.

```
score(x) = dist(x, k-th nearest neighbor)
```

- Variants: Use mean of k distances, sum of k distances.
- Time Complexity: $O(n^2)$ for brute force — use KD-tree or Ball-tree for speed.
- Sensitive to choice of k: small k = noisy, large k = misses local anomalies.
- Works well in low to moderate dimensions.

5.2 Local Outlier Factor (LOF)

LOF compares the local density of a point to the local densities of its neighbors. A point with a lower density than its neighbors gets a high LOF score.

```
reach-dist_k(A, B) = max(k-dist(B), dist(A,B)) lrd_k(A) = 1 / (sum of
reach-dist_k(A,o) for o in N_k(A)) / |N_k(A)| LOF_k(A) = (sum of lrd_k(o)/lrd_k(A)
for o in N_k(A)) / |N_k(A)|
```

- LOF ≈ 1: Point is in same density region as neighbors (normal).
- LOF >> 1: Point is in much lower density region (anomaly).
- Handles clusters of varying densities better than global methods.
- LOF score ≥ 1.5 or 2.0 is often used as a threshold.
- Variants: CBLOF, GLOF, COF (Connectivity-based Outlier Factor).

5.3 Connectivity-Based Outlier Factor (COF)

COF extends LOF by considering the connectivity structure of the neighborhood — uses chaining distances rather than direct distances, making it better for non-spherical clusters and thin elongated clusters.

5.4 Isolation Forest

An ensemble tree-based method that isolates anomalies rather than profiling normal data. It exploits the property that anomalies are few and different — they require fewer splits in a random tree to be isolated.

- Build many random Isolation Trees (iTrees) by randomly selecting a feature and a split value.
- Path length to isolate a point = anomaly score (shorter path = more anomalous).
- Anomaly Score: $s(x,n) = 2^{(-E(h(x))/c(n))}$ where $c(n)$ is average path length for n samples.
- Score → 1: highly anomalous. Score → 0.5: normal. Score < 0.5: clearly inlier.
- Time Complexity: $O(n \log n)$ — very scalable.
- Works well in high dimensions; robust to irrelevant features.
- Extended Isolation Forest (EIF) removes axis-aligned split bias.

5.5 DBSCAN for Anomaly Detection

Density-Based Spatial Clustering of Applications with Noise. DBSCAN classifies points as Core, Border, or Noise. Noise points are treated as anomalies.

eps (epsilon)	Radius defining the neighborhood of a point.

min_samples	Minimum number of points required to form a dense region (core point).
Core Point	Has at least min_samples points within eps distance.
Border Point	Within eps of a core point but fewer than min_samples neighbors.
Noise Point	Neither core nor border — treated as an anomaly.

6.1 One-Class SVM (OCSVM)

A semi-supervised method trained only on normal data. It learns a decision boundary that encloses the normal data in feature space. Points outside the boundary are anomalies.

```
Objective: maximize the margin from origin while keeping most training points within the boundary. Kernel trick maps data to higher-dimensional space for non-linear boundaries.
```

- Kernel options: RBF (most common), polynomial, linear.
- nu parameter: upper bound on the fraction of outliers (e.g., nu=0.05 for 5% anomalies).
- gamma parameter: controls influence of each training point.
- SVDD (Support Vector Data Description): fits a hypersphere around normal data.
- Limitation: Does not scale well to very large datasets.

6.2 Random Forest / Ensemble Methods

When labels are available, supervised ensembles can be trained. The class imbalance (very few anomalies) requires special handling:

- SMOTE (Synthetic Minority Oversampling) — oversample the anomaly class.
- Class weights — assign higher weight to anomaly class in loss function.
- Threshold tuning — adjust decision threshold on predicted probabilities.
- Cost-sensitive learning — different misclassification costs per class.

6.3 Angle-Based Outlier Detection (ABOD)

Measures the variance of angles between a point and all pairs of other points. Inliers have high angle variance (surrounded on all sides); outliers have low variance (far from cluster, angles concentrate in one direction).

```
ABOF(A) = Var[ / (|AB|^2 * |AC|^2) ] over all pairs B, C
```

- FastABOD approximates using only k nearest neighbors.
- Works well in high-dimensional spaces where distance methods struggle.

6.4 Minimum Covariance Determinant (MCD)

A robust estimator of multivariate location and scatter. Finds the subset of h observations (h > n/2) whose covariance matrix has the smallest determinant. Mahalanobis distances from robust estimates are used as anomaly scores.

```
Mahalanobis Distance: d(x) = sqrt[(x - mu)^T * Sigma^-1 * (x - mu)]
```

- Robust to masking effect (multiple outliers distorting the estimator).
- Assumes multivariate normality for the inlier class.

6.5 Histogram-Based Outlier Score (HBOS)

Fast unsupervised method that estimates density using histograms for each feature independently (assumes feature independence).

```
HBOS(x) = sum_i [ log(1 / hist_density_i(x_i)) ]
```

- Very fast: O(n) training, O(1) scoring per instance.
- Good baseline for high-dimensional data.

- Limitation: Feature independence assumption often violated.

6.6 Feature Bagging

Combines multiple base detectors trained on random subsets of features. Final score = average or max of base detector scores. Improves robustness and performance especially in high-dimensional data.

7.1 Autoencoders (AE)

An autoencoder is trained to compress input data into a low-dimensional latent representation (encoding) and reconstruct it (decoding). Trained only on normal data, it fails to reconstruct anomalies well — high reconstruction error signals an anomaly.

```
Architecture: Input → Encoder → Bottleneck (Latent z) → Decoder → Reconstruction
Loss: MSE(x, x_hat) = (1/n) * sum[(x_i - x_hat_i)^2] Anomaly Score = Reconstruction
Error = ||x - x_hat||^2
```

- Variants: Sparse Autoencoder, Denoising Autoencoder, Contractive Autoencoder.
- Sparse AE: Adds L1 penalty on activations — forces sparse representations.
- Denoising AE: Trained with corrupted inputs to learn robust features.
- Works for tabular data, images, text, and time series.

7.2 Variational Autoencoder (VAE)

VAE learns a probabilistic latent space — each input maps to a distribution N(mu, sigma^2) in latent space. Anomaly score combines reconstruction error and the KL divergence term.

```
ELBO Loss = E[log P(x|z)] - KL[Q(z|x) || P(z)] Anomaly Score = Reconstruction Loss +
KL Divergence
```

- Reparameterization trick: z = mu + eps * sigma, eps ~ N(0,1).
- VAE anomaly score accounts for uncertainty in the latent space.
- Beta-VAE: weighted KL term for stronger disentanglement.

7.3 LSTM-Based Anomaly Detection

Long Short-Term Memory (LSTM) networks learn temporal dependencies in sequential data. For anomaly detection, the model is trained to predict the next time step. High prediction error = anomaly.

- LSTM Autoencoder: Encoder-decoder LSTM for time-series reconstruction.
- Prediction-based: LSTM predicts t+1; anomaly = |actual - predicted| > threshold.
- Bidirectional LSTM considers both past and future context.
- Best for univariate and multivariate time series with complex patterns.

7.4 GAN-Based Anomaly Detection (AnoGAN)

Generative Adversarial Networks trained on normal data. During inference, AnoGAN finds the latent vector z that best reconstructs a query image and computes an anomaly score from the reconstruction and discriminator loss.

```
A_score = (1-lambda)*Residual_Loss + lambda*Discriminator_Loss
```

- Limitation: Requires expensive optimization at inference time.
- EGBAD, GANomaly, f-AnoGAN are faster improved variants.
- Particularly powerful for image anomaly detection.

7.5 Transformer-Based Anomaly Detection

Transformers with self-attention capture long-range dependencies. Used in time-series anomaly detection via reconstruction or prediction.

- Anomaly Transformer: Uses association discrepancy between adjacent and global attention.

- PatchTST, iTransformer, TimesNet are recent variants for time series.
- BERT-based models for NLP anomaly detection (log anomaly detection).

7.6 Deep SVDD (Support Vector Data Description)

Trains a neural network to map normal data close to a hypersphere center C in feature space. Anomaly score = distance of mapped point from center.

```
Loss = (1/n) * sum ||phi(x_i; W) - c||^2 + (lambda/2) * ||W||^2 Anomaly Score =
||phi(x; W) - c||^2
```

Unique Challenges in Time Series

- Temporal dependencies: Adjacent time points are correlated.
- Seasonality and trends must be separated from anomalies.
- Concept drift: The normal pattern itself may change over time.
- Anomalies may be contextual (e.g., spike during off-hours only).

8.1 Classical Decomposition (STL)

Seasonal-Trend decomposition using LOESS (STL) separates a time series into Trend + Seasonal + Residual components. Anomalies are detected in the residual.

```
x_t = Trend_t + Seasonal_t + Residual_t Anomaly if |Residual_t| > threshold (e.g., 3 * IQR of residuals)
```

8.2 ARIMA / SARIMA

- ARIMA(p,d,q): AutoRegressive Integrated Moving Average model.
- p = AR order (lag), d = differencing order, q = MA order.
- SARIMA(p,d,q)(P,D,Q)m adds seasonal components.
- Fit model on training data; anomalies = residuals exceeding confidence interval.
- Limitation: Assumes linear relationships and stationarity.

8.3 Prophet (Facebook/Meta)

Additive model for forecasting with daily/weekly/yearly seasonality and holidays. Anomalies are points outside the prediction interval (yhat_upper, yhat_lower).

```
y(t) = g(t) + s(t) + h(t) + epsilon_t g(t) = trend, s(t) = seasonality, h(t) = holiday effects
```

8.4 Streaming Anomaly Detection

Algorithm	Core Idea	Best For
RRCF (Robust Random Cut Forest)	Tree-based online algorithm; tracks collusive displacement	Real-time streaming
MEWMA	Multivariate EWMA control chart; tracks running mean/covariance	Process monitoring
ADWIN	Adaptive Windowing; detects concept drift in stream statistics	Concept drift detection
HSTree	Half-Space Trees; fast online density estimation	High-speed streams

8.5 Subsequence Anomaly Detection (Matrix Profile)

The Matrix Profile is a data structure that stores the z-normalized Euclidean distance between every subsequence and its nearest neighbor. Subsequences with high Matrix Profile values are the most anomalous (discord discovery).

- STAMP, STOMP, SCRIMP, STUMPY are efficient algorithms to compute Matrix Profile.

- No training required — fully unsupervised.
- Also supports motif discovery (most recurring pattern).

9.1 Confusion Matrix Components

Term	Abbreviation	Meaning
True Positive	TP	Anomaly correctly identified as anomaly
True Negative	TN	Normal point correctly identified as normal
False Positive	FP	Normal point incorrectly flagged as anomaly (Type I Error)
False Negative	FN	Anomaly missed — classified as normal (Type II Error)

9.2 Core Metrics

Precision	Formula: TP / (TP + FP) | What fraction of flagged points are true anomalies?
Recall (Sensitivity)	Formula: TP / (TP + FN) | What fraction of actual anomalies were detected?
F1-Score	Formula: 2 * Precision * Recall / (Precision + Recall) | Harmonic mean of Precision and Recall.
Accuracy	Formula: (TP + TN) / (TP + TN + FP + FN) | Overall correct classifications (misleading with imbalanced data).
AUROC	Formula: Area Under ROC Curve | Probability that anomaly scores anomalies higher than normal points. Random = 0.5, Perfect = 1.0.
AUPRC	Formula: Area Under Precision-Recall Curve | Better than AUROC for highly imbalanced datasets (few anomalies).
Matthews Correlation Coefficient (MCC)	Formula: MCC = (TP*TN - FP*FN) / sqrt[(TP+FP)(TP+FN)(TN+FP)(TN+FN)] | Balanced metric even with class imbalance.

9.3 Threshold Selection

- Fixed threshold: Set anomaly score cutoff based on domain knowledge (e.g., top 5% of scores).
- ROC-optimal: Choose threshold maximizing Youden's J = Sensitivity + Specificity - 1.
- PR-optimal: Choose threshold maximizing F1-Score on the precision-recall curve.
- Contamination rate: Set threshold so that expected proportion of anomalies matches prior knowledge.

9.4 Time-Series Specific Evaluation

- Point-adjust strategy: If any point in an anomalous window is detected, all points get credit.
- Window-based F1: Score based on anomalous time windows rather than individual points.
- NAB (Numenta Anomaly Benchmark): Rewards early detection; penalizes late detections.
- VUS (Volume Under Surface): Considers all thresholds and subsequence window sizes.

Financial Fraud Detection

- Credit card fraud: Flag unusual spending patterns (location, amount, frequency).
- Features: transaction amount, merchant category, time since last transaction.
- Techniques: Isolation Forest, Autoencoders, LSTM, GNN for transaction graphs.
- Challenge: Extreme class imbalance (fraud < 0.1% of transactions).
- Graph Neural Networks (GNN) detect fraud rings through connection patterns.

Cybersecurity & Intrusion Detection

- Network anomaly detection: Unusual traffic volume, port scanning, DDoS.
- Log anomaly detection: Rare or unexpected log event sequences.
- User Behavior Analytics (UBA): Insider threat detection.
- Techniques: LSTM for log sequences, Isolation Forest, BERT for log parsing.
- Datasets: KDD Cup 99, NSL-KDD, CICIDS2017.

Healthcare & Medical Diagnostics

- ECG anomaly detection: Arrhythmia, ST-segment changes.
- EEG anomaly: Epileptic seizure detection.
- ICU monitoring: Detecting patient deterioration early.
- Medical imaging: Tumor detection as anomaly in MRI/CT scans.
- Techniques: LSTM, CNN-Autoencoders, Transformers on waveform data.

Manufacturing & Industry 4.0

- Predictive maintenance: Detect bearing faults, motor anomalies before failure.
- Quality control: Detect defective products on assembly lines.
- Features: Vibration, temperature, current, pressure sensor readings.
- Techniques: STL, LSTM, Autoencoders, Matrix Profile.
- Datasets: MIMII, SWAT, TEP (Tennessee Eastman Process).

Key Challenges

Challenge	Description	Mitigation Strategy
Class Imbalance	Anomalies are extremely rare (< 1%)	SMOTE, cost-sensitive learning, threshold tuning, AUPRC metric
Lack of Labels	Anomalies expensive/impossible to label	Unsupervised/semi-supervised methods, active learning
High Dimensionality	Distance metrics degrade in high dims	Dimensionality reduction (PCA, t-SNE, UMAP), feature selection
Concept Drift	Normal behavior evolves over time	Online learning, sliding window, model retraining schedules
Noise vs. Anomaly	Hard to distinguish real anomalies from noise	Ensemble methods, multiple detection approaches, expert review
Scalability	Real-time detection of large-scale streaming data	Approximate algorithms (RRCF, HSTree), distributed computing
Interpretability	Black-box models hard to explain	SHAP values, LIME, attention maps, simpler models for audit

Best Practices

- Start simple: Z-Score, IQR, or Isolation Forest are often strong baselines.
- Understand your data: Visualize with t-SNE/UMAP before selecting an algorithm.
- Domain knowledge: Work with domain experts to validate detected anomalies.
- Ensemble methods: Combine multiple detectors to reduce false alarms.
- Calibrate thresholds: Use business requirements (cost of FP vs FN) to set thresholds.
- Continuous monitoring: Track model performance and retrain as data distribution changes.
- Feature engineering: Temporal features, rolling statistics, domain-specific features matter.
- Explainability: Use SHAP or feature attribution to explain why a point is anomalous.

Algorithm Selection Guide

Scenario	Recommended Algorithm(s)
Small dataset, univariate, known distribution	Z-Score, Grubbs' Test, IQR
Multivariate tabular, no labels, moderate scale	Isolation Forest, LOF, HBOS
High-dimensional data	Isolation Forest, ABOD, Deep SVDD
Univariate time series	STL + IQR, ARIMA, Prophet, CUSUM
Multivariate time series	LSTM Autoencoder, Transformer, MEWMA
Streaming / real-time data	RRCF, HSTree, ADWIN

Scenario	Recommended Algorithm(s)
Image / video anomaly detection	CNN Autoencoder, AnoGAN, SPADE
Text / log anomaly detection	BERT, LogBERT, LSTM on sequences
Graph-based data (transaction networks)	GNN (Graph Autoencoder, GAE)
Labeled data available	Random Forest, XGBoost, Neural Nets

Key Formulas at a Glance

<table>
<tr><th></th><th></th><th></th></tr>
<tr><td>Z-Score</td><td><code>z = (x - mu) / sigma</code></td><td>|z| > 3 → anomaly</td></tr>
<tr><td>IQR</td><td><code>IQR = Q3 - Q1</code></td><td>< Q1-1.5*IQR or > Q3+1.5*IQR</td></tr>
<tr><td>Mahalanobis Dist.</td><td><code>d = sqrt[(x-mu)^T Sigma^-1 (x-mu)]</code></td><td>High d → anomaly</td></tr>
<tr><td>Isolation Forest</td><td><code>s(x,n) = 2^(-E[h(x)]/c(n))</code></td><td>s > 0.5 → anomaly</td></tr>
<tr><td>LOF</td><td><code>LOF(A) = avg(lrd(neighbor)/lrd(A))</code></td><td>LOF >> 1 → anomaly</td></tr>
<tr><td>AE Reconstruction</td><td><code>score = ||x - x_hat||^2</code></td><td>High score → anomaly</td></tr>
<tr><td>VAE Score</td><td><code>score = Recon Loss + KL Divergence</code></td><td>High score → anomaly</td></tr>
<tr><td>Precision</td><td><code>TP / (TP + FP)</code></td><td>Quality of detections</td></tr>
<tr><td>Recall</td><td><code>TP / (TP + FN)</code></td><td>Coverage of anomalies</td></tr>
<tr><td>F1-Score</td><td><code>2 * P * R / (P + R)</code></td><td>Harmonic mean</td></tr>
</table>

Types Summary

Type	Level	Example
Point Anomaly	Single point	Unusually high credit card transaction
Contextual Anomaly	Context-dependent	Hot temperature in winter
Collective Anomaly	Group of points	Abnormal ECG waveform sequence

Popular Python Libraries

Library	Key Algorithms	Notes
PyOD	LOF, HBOS, ABOD, Isolation Forest, VAE, Deep SVDD, 40+ algorithms	Most comprehensive anomaly detection library
scikit-learn	Isolation Forest, One-Class SVM, LOF, DBSCAN, EllipticEnvelope	Easy integration with sklearn pipelines
STUMPY	Matrix Profile computation (STAMP, STOMP)	Best for time-series subsequence anomalies
River	RRCF, ADWIN, KSWIN	Online / streaming anomaly detection
Prophet	Additive forecasting model	Time series with seasonality + anomaly detection
TensorFlow / PyTorch	Custom Autoencoder, VAE, LSTM, Transformer models	Flexible deep learning anomaly detection

GOLDEN RULE: In anomaly detection, the cost of a False Negative (missing a real anomaly) is almost always far higher than a False Positive (false alarm). Optimize for Recall first, then tune Precision based on operational tolerance for alerts.

End of Anomaly Detection Complete Notes • Covers: Statistical, ML, Deep Learning, Time-Series, Evaluation & Applications